

Task: ROB - Robots

Lab 4. Available memory: 256 MB.

Date: 23.11.2025

Description

Bajtazar's new task is to optimally move robots to their new workstations. Bajtocja is shaped like a circle, and all robots work on the boundary. For simplicity, the boundary is divided into m sectors: $1, \dots, m$.

Currently, n robots work at locations $r_1, \dots, r_n$ ($1 \leq r_i \leq m$). It may happen that more than one robot works in some sectors.

The new work organization requires robots to be located at $t_1, \dots, t_n$ ($1 \leq t_i \leq m$) (if a sector appears k times, it means exactly k robots must be assigned there).

The cost of moving a robot from sector x to y is the cost of the shortest path along the boundary of Bajtocja:

$$\text{dist}(x, y) = \min(|x - y|, m - |x - y|)$$

A robot assignment is a bijection $f : [1, \dots, n] \rightarrow [1, \dots, n]$, where $f(i) = j$ means that robot r_i is assigned to location t_j .

The total cost of assigning all robots according to f is the sum of the transfer costs of all robots, i.e.:

$$\text{cost}(f) = \sum_{i=1}^n \text{dist}(r_i, t_{f(i)})$$

Help Bajtazar calculate the minimum cost of assigning all robots.

Input

The first line of input contains two integers separated by a single space n and m ($1 \leq n \leq 200\,000$, $1 \leq m \leq 10^9$). The second line contains n integers from the interval $[1, m]$, which are the current locations of the robots $r_1, \dots, r_n$. The third line contains n integers from the interval $[1, m]$, which are the target locations of the robots $t_1, \dots, t_n$.

Output

The program should print a single number, which is the minimum cost of assigning all robots.

Example

Input:

3 10
1 2 3
5 10 2

Output:

3